

Children's play behaviour and engagement with the schoolyard environment

Ida Lousen, Jan Arvidsen & Charlotte Skau Pawlowski

To cite this article: Ida Lousen, Jan Arvidsen & Charlotte Skau Pawlowski (2024) Children's play behaviour and engagement with the schoolyard environment, International Journal of Play, 13:4, 311-327, DOI: [10.1080/21594937.2024.2425532](https://doi.org/10.1080/21594937.2024.2425532)

To link to this article: <https://doi.org/10.1080/21594937.2024.2425532>



© 2024 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group



Published online: 19 Nov 2024.



Submit your article to this journal [↗](#)



Article views: 3464



View related articles [↗](#)



View Crossmark data [↗](#)



Citing articles: 5 View citing articles [↗](#)

Children's play behaviour and engagement with the schoolyard environment

Ida Lousen ^a, Jan Arvidsen ^b and Charlotte Skau Pawlowski ^a

^aDepartment of Sports Science and Clinical Biomechanics, World Playground Research Institute, University of Southern Denmark, Odense M, Denmark; ^bActive Living, Department of Sports Science and Clinical Biomechanics, University of Southern Denmark, Odense M, Denmark

ABSTRACT

Children's outdoor free play is declining, which correlates with decreases in physical activity and mental well-being. Given the substantial amount of time children spend in school environments, understanding the dynamics of play in the schoolyard is crucial. This study aims to identify and describe prevalent play behaviour in the schoolyard environment for pupils in first to third school year (6–9-year-olds) during break time, and to propose a typology based on these findings. In five Danish schools, observations were conducted during a total of 710 min of break time, supplemented by 12 go-along group interviews involving 57 (32 girls) pupils in first to third school year. The transcripts and field notes were analysed using thematic analysis and Gibson's theory of affordance. Five distinct types of play behaviour were identified: Rule-Bound Game Players, Imaginative Play Creators, Physical Skill Masters, Dialogue Dazzlers, and Nature Explorers. Children of the different play types perceive the schoolyard environment differently and, thus, the schoolyard affords different play behaviour. Our findings suggest that the schoolyard should offer open areas, sport facilities, loose open-ended structures, a great variety of playground structures, nature-based areas and secluded spaces to benefit all play types.

KEYWORDS

Pupils; play; schoolyard; playground; qualitative approach; affordance

Introduction

Play satisfies a multitude of vital functions in children in relation to their physical, psychological and social health, and cognitive abilities (Hyndman & Wyver, 2021), and the importance of play is underscored by its recognition in the Convention on the Rights of the Child (UNCRC, 1989). However, there has been a noticeable decrease in the time children spend outdoors in free play compared to previous generations (Frost, 2012), and this decline in outdoor free play has occurred in parallel with a reduction in children's physical activity levels (Mitchell, 2019). Research that draws on

CONTACT Ida Lousen  idalousen@health.sdu.dk  Department of Sports Science and Clinical Biomechanics, World Playground Research Institute, University of Southern Denmark, Campusvej 55, 5230 Odense M, Denmark.
 <https://www.linkedin.com/in/ida-lousen/>

© 2024 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group

This is an Open Access article distributed under the terms of the Creative Commons Attribution License (<http://creativecommons.org/licenses/by/4.0/>), which permits unrestricted use, distribution, and reproduction in any medium, provided the original work is properly cited. The terms on which this article has been published allow the posting of the Accepted Manuscript in a repository by the author(s) or with their consent.

data from 57 countries indicates that approximately one third (27–33%) of children achieve the recommended 60 min of moderate to vigorous physical activity per day (Aubert et al., 2022). Furthermore, the decline in free play time has been associated with increasing psychological stress among children (Gray, 2011). Supporting physical and mental well-being during childhood is crucial, as this developmental stage lays the groundwork for resilience against health challenges in both childhood as well as later in life (Hyndman & Wyver, 2021).

Schoolyards have been recognised as key settings for supporting children's physical activity and mental well-being (Naylor & McKay, 2009), and research has shown that playtime during school breaks accounts for up to 40% of children's daily recommended physical activity (Ridgers et al., 2006). The findings from two systematic reviews provided evidence that great variation and numerous playground facilities enhance physical activity, particularly among the least active children (Escalante et al., 2014; Ridgers et al., 2012). Further, the two reviews found that identical schoolyard interventions do not yield identical impacts on all children. For the schoolyard to reach its full potential in fostering engagement in outdoor play and physical activity, a robust understanding of children's play behaviour within this setting is crucial. This includes understanding how play practices and schoolyard environments interact, and what schoolyard facilities are essential to different types of play, seen from the children's perspective.

Several scholars have proposed different play categories, but these play categorisations have focused on play among children in general and have not considered the school environment specifically (Hughes, 2014). Other studies have categorised play behaviour based on gender differences (Boyle et al., 2003; Pawlowski et al., 2015), however, we argue that it is overly simplistic to base differentiation solely on gender when seeking to understand children's play behaviour in the schoolyard. To support a deeper understanding of healthy engagement in outdoor play among early primary school children in the schoolyard, the current study aims to identify and describe prevalent play behaviour in schoolyard environments in 6–9-year-old children during break time and to propose a comprehensive typology. The findings will offer valuable insights for practitioners tasked with designing schoolyards that cater for varied play types.

Affordance as a theoretical framework

The present study examines the relationship between children's play behaviour and their schoolyard environment through the lens of Gibson's affordance theory. The theory of affordance (Gibson, 2013) concerns the perceived relations between organisms and environments. Affordances are relational, capturing not only the characteristics of the environment but also those of the individual engaging with them. The interpretation of the possibilities afforded by an environment is influenced by the individual's size, capabilities, knowledge, what individual finds meaningful and what preferences they have in relation to interests and social interactions. At the same time, the environment influences what activity is chosen according to its design, typography, and resources (Gibson, 2013; Kyttä, 2003). In addition to affordances related to the physical environment, social affordances are defined as 'possibilities for social interaction or possibilities for action that are shaped by social practices and norms' (Carvalho, 2020). Even though

the social environment provides rich affordances for play behaviour the key focus in this study is affordances related to the physical environment.

Affordances can be divided into *potential affordances* and *actualised affordances*. Every environment has innumerable potential affordances, depending on the perceiver's qualities, such as physical fitness, motor function and imagination. Activities and actions that are actually carried out are called actualised affordances (Kyttä, 2003). In addition, scholars have proposed the concept of intended affordances, which captures the affordances of an object or environment that were intended by a designer (Burlamaqui & Dong, 2015). Thus, the theory of affordance brings attention to the diverse ways in which environments afford various play behaviours and can enrich our understanding of the possibilities children perceive in their schoolyard.

Method

The study combined observations and group interviews to get an understanding of children's play behaviour within schoolyard environments during break time. Data were collected during two-day visits at five Danish schools between October and November 2023.

Settings

The five schools had a homogeneous organisational structure of break time, which consisted entirely of free play without any organised activities. All school children had synchronised breaks twice a day at approximately 10 and 12 o'clock, playground monitors, and a policy for outdoor and mobile phone-free breaks. However, the schools varied with respect to number of children enrolled, size of schoolyard per child, minutes of break time, level of urbanisation, whether they were private or public, and regarding the range of school years offered by the school (Table 1). Additionally, the schools varied in the characteristics of the schoolyards, for example, rural and private schoolyards tended to have the most forests and thickets, while all schools equally prioritised sport courts (Figure 1).

Participants

From a group of children with parental approval, participants were purposely sampled with help from a designated teacher who knew the children well and could select children who engaged in different play activities. This approach was employed to ensure variation according to gender and play preferences to provide richer discussions (Smith & Sparkes, 2019). In total, 12 group interviews were conducted among Year 1 to Year 3 pupils aged 6–9 years old, encompassing 57 participants (32 girls) (Table 2).

Table 1. School characteristics.

School	Children enrolled (n)	Schoolyard size per child (m ²)	Minutes of break time per day	Rural/urban	Private/public	School year
1	266	56	50	Rural	Public	0–6
2	230	24	60	Urban	Private	0–9
3	100	177	60	Rural	Public	0–6
4	429	32	60	Urban	Public	0–9
5	358	44	45	Urban	Public	0–6

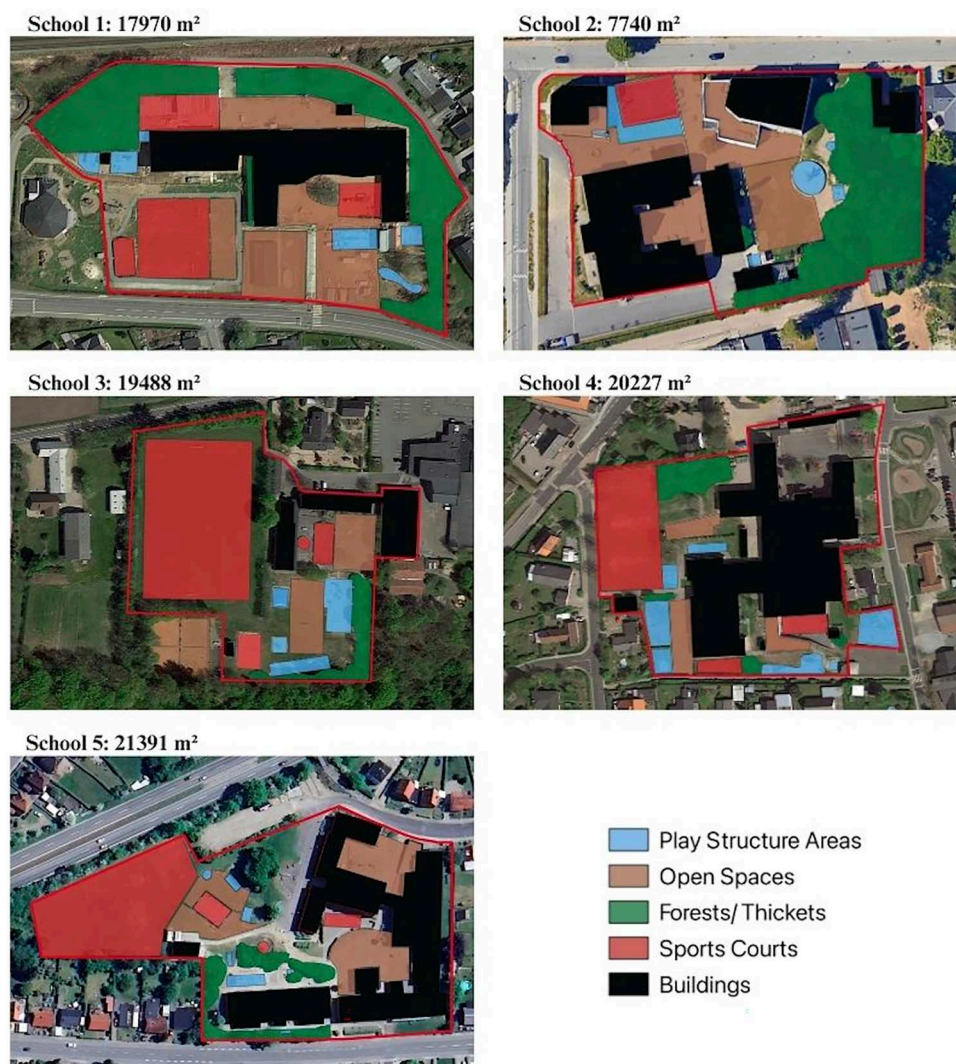


Figure 1. Schoolyard characteristics.

Data collection

A total of 710 min of break time were observed by two researchers (IL and CSP). The observations were open-ended with no observation guide; however, they had a focus on the children's play behaviour and engagement with the physical environment. Children's play behaviour was observed at different locations in the schoolyard (e.g. football court, sandpit, green area). Each location was observed for approximately 15–20 min. Occasionally, the observers asked the children questions about what they were doing but generally they did not interact. The observations were documented using ethnographic fieldnotes, with a focus on capturing detailed descriptions of play situations (Emerson et al., 2011).

The group interviews were conducted during classes between breaks and followed a two-phase procedure: In the first phase, a go-along group interview was conducted, in

Table 2. Participants in group interviews.

Interview group	School	School year	Number of participants (gender)
1	1	1	5 (2B + 3G)
2	1	2	5 (2B + 3G)
3	1	3	4 (2B + 2G)
4	2	1	6 (3B + 3G)
5	2	2	6 (1B + 5G)
6	2	3	5 (4B + 1G)
7	3	1	4 (1B + 3G)
8	4	1	4 (2B + 2G)
9	4	2	5 (3B + 2G)
10	4	3	5 (2B + 3G)
11	5	1	4 (2B + 2G)
12	5	3	4 (1B + 3G)
Total	12	Year 1: 23 Year 2: 16 Year 3: 18	57 (25 boys and 32 girls)

which the same two researchers (IL + CSP) served as moderators and were guided around the schoolyard by small groups of children (see Table 2). The children showed the moderator their usual play spaces and elaborated on the games they typically engaged in. The go-along group interviews encompassed all areas of the schoolyard that the children recalled interacting with. Prompts while walking with the children included: ‘What do you do during break time?’ ‘Why do you enjoy that activity?’, and ‘How do you use this structure?’ This go-along approach was chosen to help the children recall memories of their play during break time (Kusenbach, 2003). The second phase happened immediately after the go-along interview, when the moderator and the small group of children moved from the schoolyard to a meeting room to have a focus group discussion. The children were introduced to 19 photos brought by the researchers depicting common play activities (e.g. playing tag, looking at insects, sitting alone on a bench, walking, and talking, and fencing with sticks). Given the young age of the children, we found it beneficial to employ visual aids to evoke thoughts of their play activities and prompt discussions on play behaviour in their schoolyard. Similar research has been done by Veitch and colleagues (2021). In the present study, the children were asked to select the photos showing what they could do in the schoolyard and to explain who did what and how. The children were encouraged to describe whether other play activities than those shown on the photos were practiced in the schoolyard. Thus, supported by the photos, the researchers and children had an exploratory and verifying discussion leading back to the topics presented during the go-along group interview. The go-along group interview and focus group discussion lasted approximately 45 min in total and were video recorded to capture the dialogue and to identify the person speaking. Subsequently, the interviews were transcribed, and the schools and participants were pseudonymised.

Data analysis

The purpose of the analysis was to identify a typology of play behaviour, drawing equally on fieldnotes and interview transcripts. The analysis followed Braun and Clark’s six phases of thematic analysis, with the aim of systematically uncovering patterns of meaning across the dataset (Braun et al., 2016). The first steps involved familiarisation

with and initial systematic coding of all field notes and interview transcripts in NVivo 1.7.1 to ensure a comprehensive understanding of all data. Following this, codes were organised based on patterns in the play behaviour and how the children used the schoolyard environment. Based on similarities and differences, a schoolyard play typology began to emerge, and further development of the types (including refinement, naming, and writing) was performed as a recursive and reflexive process. Data were analysed inductively to prevent the imposition of theory that might obscure interpretations of the actual play occurring in the schoolyards. Lastly, after having identified different types of play behaviour across the data, the relation between children and the physical environment was analysed using perspectives from Gibson's theory of affordance (Gibson, 2013).

Ethics

Approval for the study was granted by Research Ethics Committee at the University of Southern Denmark (23-60290). In addition, the study was approved by the principals of the five schools, and written informed consent was obtained from parents. On the day of the group interview, the children were also informed about the study and asked if they wished to participate. The children were informed that they could withdraw from the study at any time and without providing a reason. Five of the children with parental approval withdrew from the study before interview.

Results

Five types of play behaviour – Rule-Bound Game Players, Imaginative Play Creators, Physical Skill Masters, Dialogue Dazzlers, and Nature Explorers – were present across all schoolyards among substantial numbers of children and will be elaborated on in the following. Each type should be seen as a description of common play behaviour rather than specific children's actions and preferences. Although some children seem to have preference for a specific type of play behaviour, it is important to note that children engage in different types of play behaviour during the day and over time. Thus, the ambition of the study is to produce and present a typology of common schoolyard-based play behaviour – not to classify children. Each type of play behaviour described below exists independently with its own unique characteristics. However, minor overlaps between the different play types exist.

Rule-Bound game players

A substantial cohort of children exhibited a keen preference for physical activity, involving actions such as running, throwing, or kicking a ball. These children often engaged in games with predefined rules. Typically, games within this type included football (soccer), bat-and-ball games, tag, and hide-and-seek. Tag games and most ball games engaged both girls and boys, while football seemed to be played primarily by boys.

The children playing these kinds of games were driven by competition, encompassed by a desire to run faster, score goals, or be the last to be tagged or found. Moreover, children immersed in these play activities demonstrated remarkable resilience:

A large player dribbles towards a way smaller goalkeeper. At that very moment, he collides with the goalkeeper, causing both to tumble over. The ball manages to find its way into the goal. Unaffected, the goalkeeper quickly rises, retrieves the ball, and proceeds to take a goal kick. (Field note, School 2)

This quote exemplifies that Rule-Bound Game Players display an absence of apprehension about being struck by a ball, readily engaging in collaborative tumbles without concern for bruises or soiled clothing. Since competition seemed important for the Rule-Bound Game Players, it was also important that the participants abided by the rules of the game. Failure to comply with the rules often resulted in frustration, conflicts or even exclusion:

A boy approaches the playground monitor and tells her that the others have kicked him out of the game. The other children explain that it was because he had continued running even after being declared 'dead' several times. Some of the boys from the game agree with the monitor that he can join again if he promises to follow the rules. (Field note, School 4)

Discussions and conflicts frequently arose within these rule-based games, illustrating the complicated social dynamics within this type. Prior to the data collection, School 1 had experienced numerous conflicts related to football, prompting the management to implement a 'ball-free week', which revealed yet another interesting type of behaviour. During this time, the children who usually played football shifted their focus to a hide-and-seek game. This suggests that while the children within this type could easily modify their preferred activities, they usually leaned towards activities that had common elements, such as playing by rules, being competitive and physically active.

The games played by Rule-Bound Game Players were played within larger group settings, most often in open areas or sport courts. Most structures were utilised as intended, indicating that the play behaviour of Rule-Bound Game Players is supported by the intended affordances of the environment. For example, football pitches were used for football, open spaces were utilised for running, and bats were used to hit balls. Only games like hide-and-seek required a more creative use of the environment. Children engaging in hide-and-seek games sought suitable hiding spots, often utilising structures intended for other purposes (e.g. climbing frames and small tree huts).

Imaginative play creators

Numerous children transitioned from reality to an imaginative realm through play. Children engaging in these types of games exhibited distinct creativity and spontaneously invented characters and rules, continuously evolving their play. The children played in small groups, typically consisting of 2–5 children, predominantly separated into groups of boys or girls. Drawing inspiration from a mix of movies, books, and the tangible world around them, these children took on traditional gender roles, with girls often engaging in domestic duties like baking or caring for children, or assuming roles such as fairies or witches:

In a thicket, two girls walk around with sticks in their hands. One girl picks some white berries and throws them up into the air and says that they are glitter and rainbow hearts falling over them. She points her stick towards a branch on a bush and shouts 'sprout magic', explaining to the other girl that leaves will now appear on the branch. (Field note, School 2)

On the other hand, boys frequently took on the roles of cops, robbers, spies, and warriors in warlike scenarios. Moreover, boys and girls engaged differently in imaginative play, with girls' play often being calmer and more peaceful and boys' play being energetic and more physical:

One boy walks around making threatening sounds, while the other two boys try to overpower him in different ways. One wrestles him down, gets attacked, and is brought down in a hold, after which the third boy saves the downed boy by touching him and saying 'life, life'. (Field note, School 1)

In contrast to observations of the Rule-Bound Game Players, we did not observe conflicts within this type. Children with play behaviour adhering to the Imaginative Play Creators stated that they avoided conflicts in their play because they were not bound by defined rules:

Mia: Yes, you can completely decide what you want, you can be a family or have magic powers, it's entirely up to you. For example, you can be a family and then someone says, 'I want to have magic powers', and then you have them [magic powers], and there's no one to say you can't or you're not allowed.

Moderator: So, you don't argue when you come up with the game?

Mia: It's a game everyone can join in, and you can do whatever you want. (School 1, Year 3)

When children experienced unfair treatment or perceived an activity as uninteresting, they modified the play:

A boy lies quietly on the ground nearby, but shortly after, he returns to two other boys playing cops and robbers and says, 'I had a healing, so I could come back to life again'. (Field note, School 5)

The children explored every corner of the schoolyard, from open areas to thickets and play structures. They shifted between traversing the schoolyard, discovering various potential affordances in the environment, and staying in selected spots, such as up a tree or beside the sandpit, where they lingered for a while. Resourceful in their play, they made use of whatever was available, for example lunchboxes, branches, walls, or swings, and engaged with these facilities in ways that far exceeded the intended affordances. Natural and built structures shared numerous similarities for children of this type, as they viewed them as environments with several and fluctuating affordances. Nevertheless, play with natural elements seemed particularly prevalent, and it appeared that children perceived more potential affordances in natural materials:

Ava: Sometimes we also play McDonald's in that tree over there.

Moderator: Why do you choose that particular tree to play McDonald's in?

Ava: Because it's a big tree with lots of branches where you can play, for example, there are two branches that go like this, and you can pretend to make milkshakes (...).

Oliver: And sometimes we pretend you can drive up here.

Ava: Yes, and then you can take orders here and here you squeeze ice and here you can sit and make burgers.

Oliver: And here's the gear stick. (School 1, Year 3)

The dynamic structures and randomness inherent in the composition of the tree, like other natural features in the schoolyard, seemed to be particularly effective in stimulating the children's imagination.

Physical skill masters

Another type of play behaviour encompassed the behaviour of children actively engaging with structures such as climbing frames, trampolines, swings, and climbing trees. The children typically perceived the affordances of these structures as intended by the designers, meaning that they climbed on structures designed for climbing and swung on structures designed for swinging. The different structures were used by small groups, usually comprising 2–5 children. This play behaviour was prevalent among groups of both boys and girls, with same-gender groups being most common.

Children who exhibited the Physical Skill Master type of behaviour utilised their motor skills and often possessed well-developed gross motor abilities, showcasing their proficiency in hanging, climbing, jumping, balancing, coordinating limbs, and challenging their vestibular senses. The requirement of motor skills excluded children who either lacked the courage to embrace the challenge or did not possess the necessary skills:

Three boys arrive at the climbing wall. Two of the boys climb up the wall, while the third boy stands on the ground and kicks the wall, occasionally holding onto the lower grips without making any attempt to ascend. (Field note, School 1)

The games played depended on available structures in the schoolyard, thus varied from school to school. However, common to all schools was that the provided structures served as the context, and children filled the context with content of diverse and unique character, like rules and stories. This implies an overlap with other types of play behaviour; however, the primary focus of this play was on mastering physical skills, and many of the children engaging with this play behaviour were driven by their eagerness to ‘accomplish something’:

Lily: When we were in Year 2, I feel like some of the boys were trying to show off a bit.
 Isla: So, they could jump over five buckets, and then they could, like, jump 10 times around or something like that. (School 5, Year 3)

This quote suggests that the boys were driven by their ability to master challenging skills and gain admiration from their peers. The boys focused more on the physical endeavour itself rather than on competition. Although Physical Skill Masters also engaged with competitive elements, the competitions were organised in a gentler manner compared to those of the Rule-Bound Game Players and ensured that all participants had positive experiences:

Olivia: We do it [gymnastics] quite often together because we think it is fun and enjoyable to do it with others.

Moderator: Is it okay to not win the competition too? Or is it more fun to be the one who wins?

Olivia: It is also okay not to win, but you can get up to 1000 [points], and we often get that, so sometimes we also do our worst so that they can get some points too. (School 1, Year 2)

This quote exemplifies the idea of modifying competition to ensure inclusivity and positive experiences, whereby participants prioritised collective enjoyment and fairness over individual victory to keep the game going. Another prevalent trend among children of this type was the children’s inclination to seek excitement or a ‘thrill’: ‘We often climb trees, it’s so much fun to climb in trees. It’s a bit dangerous because you must stand

on this thin branch' (Emily, School 2, Year 2). These actions appear to elicit a sensation of excitement, and the children seemed drawn to the thrill of the slightly dangerous situations in their play.

In this type of play, a challenge arose when there was competition for access to the same play structures:

Four girls are swinging on a bird's nest swing, and two boys stand 2–3 meters away looking displeased. The girls shout in unison, 'Girls are the best!' repeatedly. The playground monitor approaches the boys, after which they go and sit in the climbing frame, glancing at the girls who are swinging loudly and energetically. (Field note, School 1)

The scarcity of certain structures triggered competition and territorial behaviour among the children, ultimately impacting their play experiences and interactions with each other.

Dialogue dazzlers

In some situations, conversation was prioritised over play. By forming small groups of 2–3 individuals, the children created intimate social settings in which they occupied themselves in dialogue. There was a noticeable inclination towards same-gender interactions during these conversations, with more groups comprised of girls. They engaged in discussions on various topics, with recurring themes including recounting experiences from weekends, holidays, or plans for celebrating specific occasions. Some conversations revolved around niche interests, such as Pokémon cards or fantasy worlds. Additionally, a significant number of girls chatted about topics related to boys.

In this type, conversation emerged as the primary activity, but the children also engaged in secondary pursuits, such as swinging:

Two boys stand in a bird's nest swing, synchronously with one leg forward, pushing their weight down on the leg to gain momentum. The swing almost reaches horizontal. While the boys swing, they talk about various topics. They discuss carving pumpkin heads, which they did during the holidays. The conversation evolves into a deeper discussion, where one boy reveals that he knows someone who has just died from an illness. (Field note, School 2)

Other activities while conversing included strolling, sitting, climbing, or interacting with nature, for example, by using a stick to dig in soil. The children distanced themselves from the chaos in the schoolyard because, as one girl explained, '*It [the schoolyard] feels very tight, and very noisy*' (Poppy, School 1, Year 1). Distancing might involve sitting in a tree or a tower or finding refuge in a hidden area among some bushes. Another way to isolate themselves was by walking arm in arm, which was done solely by girls:

Two girls arrive at the forest arm in arm. They stand by a tree for a while, scratching the tree trunk as they agree to play together after school. They continue walking through the forest, chatting about what they did over the weekend. (Field note, School 1)

Regardless of the distancing method employed, the children conveyed that they belonged to an exclusive group, which was seemingly closed off to additional participants.

Dialogue Dazzlers often perceived other affordances than those obviously intended in the environment. For instance, a climbing tower did not inherently encourage climbing

for these children; rather, it invited them to sit quietly and engage in conversation. The children were particularly drawn to features of the environment that offered some degree of refuge. Similarly, the same principle applied to other structures, as these children sought out peaceful spaces.

Nature explorers

Some children had a special interest in the real, living world around them, expressed by one boy in this way: *'I like to explore the forest'* (Jacob, School 2, Year 3). The activity of 'exploring' was for Nature Explorers a task in itself. The natural environments found in the schoolyards often consisted of only a couple of trees and a few square metres of vegetation. Despite this, the children still engaged in activities that reflected their curiosity and interest in their natural surroundings. Searching for animals was a particularly common activity that was observed:

Moderator: Why do you find animals?

Sophia: I think it's nice to walk around in nature.

Moderator: What do you do with the animals?

Sophia: We pet them, and then when we don't want them anymore, we release them back into nature.

Moderator: What kind of animals do you find?

Sophia: Ladybirds and forest snails and things like that. (School 1, Year 1)

It appeared that there were two main reasons for the children's interest in finding these animals. Firstly, walking around in nature-like surroundings and exploring created a sense of comfort and enjoyment for them. Secondly, these children exhibited a desire to care for animals, which was evident in their interactions with the creatures. Despite the children being engaged in other activities, the presence of animals could divert their attention and spark their curiosity:

'Look, there's a ladybird', she says. She pokes the ladybird in the water for a bit, while the other girl continues playing, and the girl with the ladybird stands up again and disappears back into the game. (Field note, School 2).

Similar scenarios also unfolded when people passed by the schoolyard with a dog and when a cat appeared. The children became completely absorbed in the presence of these animals, forgetting about their previous activities. Another method the children employed to nurture their interest was by imitating animals:

Three girls gather acorns from the forest floor and hide them at the base of a tree in a small hollow by some tree roots. They try to open the small acorns by jumping on them. (Field note, School 1)

The girls were collecting acorns, mimicking the behaviour of squirrels preparing for winter. This behaviour not only reflected their interest in animals, but also highlighted their ability to learn from and imitate the natural world around them. Other girls pretended to be horses, rabbits, and dogs in the same way in other schoolyards.

The activities were typically undertaken by groups comprising 2–3 children in either same-gender or mixed-gender groups. The Nature Explorers perceived the schoolyard environment by adopting the perspective of actual animals. When they looked around the schoolyard, the environment afforded residential shelters and

food, whereas to other children it afforded items such as climbing frames, hiding spots, and weapons.

Discussion

A play typology representing common play behaviour and engagement with schoolyard environments among Year 1 to Year 3 children was identified. The play typology consists of five play types: (1) Rule-Bound Game Players, who played competitive and energetic games; (2) Imaginative Play Creators, who explored fantasy worlds; (3) Physical Skill Masters, who were excited about thrill and bodily mastery; (4) Dialogue Dazzlers, who engaged in conversation; and (5) Nature Explorers, who embraced adventure and wild-life. In a previous study, six distinct play types in the schoolyard were identified among Year 4 pupils (10-11-year-olds) (Pawlowski et al., 2015). In the study, children's play revolved around similar activities to those in the current study, such as ball games, tag games, imaginary play, and conversation. However, there were notable differences between the two studies in the prevalence of certain activities. Among the Year 4 pupils, football emerged as a dominant activity in which skilled players ranked highly in the schoolyard hierarchy. Conversely, boys with an interest in fantasy were marginalised and labelled as 'nerds' by the children (Pawlowski et al., 2015). In the current study of younger children, imaginary play was more prevalent and socially acceptable than among the Year 4 pupils. The difference in play preferences in the schoolyard among different age groups calls for variety in the schoolyard environment. The theory of affordance suggests that affordances become available as a result of both the properties of the environment and the perceptions of the organism, and that affordances prompt behaviour (Gibson, 2013). Imaginative Play Creators and Nature Explorers were more frequent in schools 1 and 2. As shown in Figure 1, schools 1 and 2 have more forest and thicket coverage compared to schools 3, 4, and 5, which indicates that the design of the schoolyards inevitably affects children's play behaviour. Studies have found that the design of many schoolyards favours children interested in sport activities, for example football, while children inclined to engage in other activities are favoured less (Pawlowski et al., 2015; Ridgers et al., 2010). In accordance with this, a substantial area of the schoolyards investigated in the current study were laid out as sport courts and open spaces, offering extensive and intended opportunities for Rule-Bound Game Players. This allocation of space tends to leave fewer intended opportunities for the Imaginative Play Creators, Dialogue Dazzlers, and Nature Explorers, who favoured nature-based environments, despite the limited presence of these in schoolyards. In line with this finding, Amicone and colleagues found that built environments were predominantly used for games with rules, while significantly more explorative play behaviour, active conversations, and rough-and-tumble play were observed in natural environments (Amicone et al., 2024). Other studies conducted in primary schools have also found that natural elements on school grounds 'expand the children's play repertoire' by engaging them in less rule-bound play and more unorganised 'free' and creative play compared to the conventional schoolyard (Dyment et al., 2009; Moore & Wong, 1997). It seems, however, not to be the natural environment itself that attracts the Imaginative Play Creators and Dialogue Dazzlers as it does for the type of Nature Explorers. Dialogue Dazzlers prioritised nature-based environments because these spaces were often more peaceful and secluded, and for

Imaginative Play Creators the natural environments seemed attractive due to the presence of loose parts, such as branches and berries. Other secluded spaces in the schoolyard were also used by the Dialogue Dazzlers, such as the top of a climbing frame. The necessity for secluded spaces in the schoolyard among upper primary school children has also previously been addressed (Pawlowski et al., 2014, 2018).

In accordance with previous studies acknowledging the necessity of structures motivating risky play (Brussoni et al., 2015), challenging and thrilling structures in the schoolyard, such as climbing frames, trampolines, and swings, seem important for the Physical Skill Masters. Additionally, the Rule-bound Game Players favoured structures such as football goals for their play. In line with this, some studies have found positive correlations between well-equipped schoolyards and physical activity levels (Dyment et al., 2009; Ridgers et al., 2010), and other studies have identified the lack of facilities as a barrier to schoolyard play (Martínez-Andrés et al., 2017; Pawlowski et al., 2018). Further, conflicts over structures tend to arise among Physical Skill Masters due to scarcity of spaces. The provision of a variety of facilities has shown to promote inclusion by accommodating many children playing together or alongside each other (Wenger et al., 2024), which points to the importance of having sufficient play structures in the schoolyard, thereby enhancing the play opportunities for more children. Such findings indicate that physical and social affordances interact (Carvalho, 2020; Gibson, 2013) and conspire to shape play behaviour. This might suggest that the benefits of providing a variety of facilities could be amplified by modifications of play norms and culture.

It appears that there are significant differences in the number of actualised affordances that children find in various environments. Environments with the fewest different actualised affordances include sports courts, since they are primarily used for ball games as intended. With play structures, children actualise more varied affordances, by extending and enriching these intended affordances (e.g. swinging and sliding) with additional, unintended affordances through storytelling and inventing new games. It seems that the environments which afford the most multifarious play behaviour are environments that contain loose parts, as these can accommodate a wide variety of imaginative and explorative play behaviour. Studies have shown that loose objects can foster creativity, cooperation among peers, encourage ingenuity (Watkins, 2020), and promote physical activity (Ridgers et al., 2012). This potential also applies to balls (Zask et al., 2001), which makes the availability of loose parts interesting for both Rule-Bound Game Players and Imaginative Play Creators. Research shows that children exhibit no specific preferences for natural or synthetic loose objects (Watkins, 2020). However, our findings suggest that children prefer playing with natural loose objects. In the five investigated schoolyards, a limited number of man-made loose objects were provided, which potentially contributed to the children's strong inclination to play with natural elements such as sticks, berries, and leaves.

Strengths and limitations

The equally use of two complementary methods strengthened the current study by enriching the data and improving the credibility of the results (Darbyshire et al., 2005). The consistency of findings from the children across the five different

schools suggests that the results have a broad application across a variety of schoolyard environments; however, it is important to be aware that the selection of both schools and participants introduces some degree of bias (Robinson, 2014). The data were collected in Denmark in the autumn season, which limits the generalisability of the findings to other cultural and seasonal contexts. Further, according to the theory of affordance, the environment influences play behaviour. It is possible that different schoolyard facilities might have elicited play behaviour not identified in this study. The selection process may have influenced data in the sense that the selection was limited by the teachers' understanding of what constitutes different play behaviour. Nevertheless, this procedure was more feasible than extensive screening of children's play behaviour and based on knowledge stemming from observations in the schoolyards, we assess that a diverse representation of play behaviour was present in the interview groups. Moreover, the predetermined set of 19 photos can potentially lead to biased findings, as photos 'invite' children to mainly discuss activities shown on the pictures. However, we strategically scheduled the activity involving photos to occur last and encouraged discussions on activities beyond those depicted in the 19 photos. Last, the interviews only included the children's perspectives. The perspectives of teachers, school management, and parents may differ from the children's viewpoints.

Conclusion

The identification and description of five types of play behaviour, derived from early primary schoolchildren's play behaviour within the schoolyard environment, offer valuable insights into how schoolyard design can be optimised to cater for all types of playing. Children from each type of play behaviour perceive the schoolyard differently and, thus, actualise different affordances of the environment. This underscores the necessity for tailored interventions that are informed by deep and nuanced analysis of prevalent play behaviour in schoolyard environments to promote healthy engagement in outdoor play across all types of playing. Based on the findings, we recommend designing schoolyards that incorporate open areas, sport facilities, loose parts, open-ended structures, a great variety of playground equipment, nature-based areas, and secluded spaces to secure variety that accommodates all kinds of play behaviour.

Acknowledgements

We thank all the participating children and their parents for taking part in our study. We also thank the school staff for their assistance in organising the data collection.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

This study was financially supported by KOMPAN, a large international playground equipment company. KOMPAN had no influence on the methods, results, or conclusions.

Notes on contributors

Ida Lousen, MSc in Sport Science and Health, works as a research assistant affiliated with the World Playground Research Institute within the Department of Sports Science and Clinical Biomechanics, University of Southern Denmark. Her focus lies in conducting theoretically informed qualitative and quantitative research concerning children and adolescents engaging in outdoor activities.

Jan Arvidsen is associate professor in the research unit Active Living at University of Southern Denmark. He does theoretically informed qualitative and quantitative research aiming to provide adequate physical, social, and managerial settings for outdoor recreation and active outdoor living. His research has a particular focus on (1) Children's experiences and use of nature, (2) Motives, preferences and behaviours in outdoor recreation and outdoor active living, (3) Co-creation and volunteering in outdoor recreation, (4) Social conflict in outdoor recreation, and (5) Motivation, adherence, and volunteering in peer-led walking.

Charlotte Skau Pawlowski is associate professor in the World Playground Research Institute at University of Southern Denmark. She does research on the relation between human behaviour and the physical and social environment. Charlotte's primary research focus revolves around creating inclusive playgrounds that cater to children with diverse abilities and backgrounds. At the heart of her research is the belief that children should have a say in the design process.

ORCID

Ida Lousen  <http://orcid.org/0009-0005-0843-8781>

Jan Arvidsen  <http://orcid.org/0000-0001-5352-3567>

Charlotte Skau Pawlowski  <http://orcid.org/0000-0003-1407-613X>

References

- Amicone, G., Collado, S., Perucchini, P., Petrucci, I., Ariccio, S., & Bonaiuto, M. (2024). Green versus grey break: Children's place experience of recess-time in primary schools' natural and built area. *Children's Geographies*, 15–29. <https://doi.org/10.1080/14733285.2023.2207118>
- Aubert, S., Barnes, J. D., Demchenko, I., Hawthorne, M., Abdeta, C., Abi Nader, P., Adsuar Sala, J. C., Aguilar-Farias, N., Aznar, S., Bakalár, P., Bhawra, J., Brazo-Sayavera, J., Bringas, M., Cagas, J. Y., Carlin, A., Chang, C. K., Chen, B., Christiansen, L. B., Christie, C. J., ... Paudel, S. (2022). Global Matrix 4.0 physical activity report card grades for children and adolescents: Results and analyses from 57 countries. *Journal of Physical Activity & Health*, 19(11), 700–728.
- Boyle, D. E., Marshall, N. L., & Robeson, W. W. (2003). Gender at play. *American Behavioral Scientist*, 46(10), 1326–1345. <https://doi.org/10.1177/0002764203046010004>
- Braun, V., Clarke, V., & Weate, P. B. (2016). Using thematic analysis in sport and exercise research. In B. Smith & A. C. Sparkes (Eds.), *Routledge handbook of qualitative research in sport and exercise* (pp. 191–205). Routledge. <https://doi.org/10.4324/9781315762012>
- Brussoni, M., Gibbons, R., Gray, C., Ishikawa, T., Sandseter, E. B., Bienenstock, A., Chabot, G., Fuselli, P., Herrington, S., Janssen, I., Pickett, W., Power, M., Stanger, N., Sampson, M., & Tremblay, M. S. (2015). What is the relationship between risky outdoor play and health in children? A systematic review. *International Journal of Environmental Research and Public Health*, 12(6), 6423–6454. <https://doi.org/10.3390/ijerph120606423>
- Burlamaqui, L., & Dong, A. (2015). *The identification of perceived intended affordances*. IASDR Congress. <https://doi.org/10.13140/RG.2.1.4555.7209>
- Carvalho, E. (2020). Social affordance. In J. Vonk & T. Shackelford (Eds.), *Encyclopedia of animal cognition and behavior* (pp. 6497–6500).

- Darbyshire, P., MacDougall, C., & Schiller, W. (2005). Multiple methods in qualitative research with children: More insight or just more? *Qualitative Research*, 5(4), 417–436. <https://doi.org/10.1177/1468794105056921>
- Dymont, J., Bell, A., & Lucas, A. (2009). The relationship between school ground design and intensity of physical activity. *Children's Geographies*, 7(3), 261–276. <https://doi.org/10.1080/14733280903024423>
- Emerson, R. M., Fretz, R. I., & Shaw, L. L. (2011). *Writing ethnographic fieldnotes* (2nd ed.). University of Chicago Press.
- Escalante, Y., García-Hermoso, A., Backx, K., & Saavedra, J. M. (2014). Playground designs to increase physical activity levels during school recess. *Health Education & Behavior*, 41(2), 138–144. <https://doi.org/10.1177/1090198113490725>
- Frost, J. (2012). The changing culture of play. *International Journal of Play*, 1(2), 117–130. <https://doi.org/10.1080/21594937.2012.698461>
- Gibson, J. J. (2013). *The ecological approach to visual perception* (Classic ed.). Psychology Press. <https://doi.org/10.4324/9780203767764>
- Gray, P. (2011). The decline of play and the rise of psychopathology in children and adolescents. *American Journal of Play*, 3(4), 443.
- Hughes, B. (2014). Playtypes and the playwork menu. In B. Hughes (Ed.), *Evolutionary playwork* (pp. 96–133). Taylor & Francis Group.
- Hyndman, B., & Wyver, S. (2021). *Outdoor recreation within the school setting: A physiological and psychological exploration* (outdoor recreation-physiological and psychological effects on health. IntechOpen. <https://doi.org/10.5772/intechopen.91900>
- Kusenbach, M. (2003). Street phenomenology. *Ethnography*, 4(3), 455–485. <https://doi.org/10.1177/146613810343007>
- Kyttä, M. (2003). *Children in outdoor contexts. Affordances and independent mobility in the assessment of environmental child friendliness*. Helsinki University of Technology].
- Martínez-Andrés, M., Bartolomé-Gutiérrez, R., Rodríguez-Martín, B., Pardo-Guijarro, M. J., & Martínez-Vizcaíno, V. (2017). “Football is a boys’ game”: children’s perceptions about barriers for physical activity during recess time. *International Journal of Qualitative Studies on Health and Well-Being*, 12(1), 1–9. <https://doi.org/10.1080/17482631.2017.1379338>
- Mitchell, J. (2019). Physical inactivity in childhood from preschool to adolescence. *ACSM’S Health & Fitness Journal*, 23(5), 21–25. <https://doi.org/10.1249/FIT.0000000000000507>
- Moore, R. C., & Wong, H. H. (1997). *Natural learning: the life of an environmental schoolyard*. Creating environments for rediscovering nature’s way of teaching. Mig Communications.
- Naylor, P. J., & McKay, H. A. (2009). Prevention in the first place: Schools a setting for action on physical inactivity. *British Journal of Sports Medicine*, 43(1), 10–13. <https://doi.org/10.1136/bjsm.2008.053447>
- Pawlowski, C. S., Ergler, C., Tjørnhøj-Thomsen, T., Schipperijn, J., & Troelsen, J. (2015). ‘Like a soccer camp for boys’. *European Physical Education Review*, 21(3), 275–291. <https://doi.org/10.1177/1356336X14561533>
- Pawlowski, C. S., Schipperijn, J., Tjørnhøj-Thomsen, T., & Troelsen, J. (2018). Giving children a voice. *European Physical Education Review*, 24(1), 39–55. <https://doi.org/10.1177/1356336X16664748>
- Pawlowski, C. S., Tjørnhøj-Thomsen, T., Schipperijn, J., & Troelsen, J. (2014). Barriers for recess physical activity: A gender specific qualitative focus group exploration. *BMC Public Health*, 14(1), 639–639. <https://doi.org/10.1186/1471-2458-14-639>
- Ridgers, N. D., Fairclough, S. J., & Stratton, G. (2010). Variables associated with children’s physical activity levels during recess: The A-CLASS project. *International Journal of Behavioral Nutrition and Physical Activity*, 7(1), 74. <https://doi.org/10.1186/1479-5868-7-74>
- Ridgers, N. D., Salmon, J., Parrish, A. M., Stanley, R. M., & Okely, A. D. (2012). Physical activity during school recess. *American Journal of Preventive Medicine*, 43(3), 320–328. <https://doi.org/10.1016/j.amepre.2012.05.019>
- Ridgers, N. D., Stratton, G., & Fairclough, S. J. (2006). Physical activity levels of children during school playtime. *Sports Medicine*, 36(4), 359–371. <https://doi.org/10.2165/00007256-200636040-00005>

- Robinson, O. C. (2014). Sampling in interview-based qualitative research: A theoretical and practical guide. *Qualitative Research in Psychology*, 11(1), 25–41. <https://doi.org/10.1080/14780887.2013.801543>
- Smith, B., & Sparkes, A. C. (2019). Interviews. Qualitative interviewing in the sport and exercise sciences. In B. Smith & A. C. Sparkes (Eds.), *Routledge handbook of qualitative research in sport and exercise* (pp. 103–123). Routledge.
- UNCRC. (1989). *Convention on the Rights of the Child*. UNICEF. Retrieved February 15 from <https://www.unicef.org/child-rights-convention/convention-text>
- Veitch, J., Ball, K., Flowers, E., Deforche, B., & Timperio, A. (2021). Children’s ratings of park features that encourage park visitation, physical activity and social interaction. *Urban Forestry & Urban Greening*, 58, 126963. <https://doi.org/10.1016/j.ufug.2020.126963>
- Watkins, G. L. (2020). An Action Research Study of Creativity and Physical Activity at a Southern Preschool Through Natural and Synthetic Loose Parts for Traditional Playgrounds [Ed.D., Northcentral University]. ProQuest Dissertations & Theses Global.
- Wenger, I., Lynch, H., Prellwitz, M., & Schulze, C. (2024). Children’s experiences of playground characteristics that contribute to play value and inclusion: Insights from a meta-ethnography. *Journal of Occupational Science*, 405–432. <https://doi.org/10.1080/14427591.2023.2248135>
- Zask, A., van Beurden, E., Barnett, L., Brooks, L. O., & Dietrich, U. C. (2001). Active school playgrounds—myth or reality? Results of the “move It groove It” project. *Preventive Medicine*, 33(5), 402–408. <https://doi.org/10.1006/pmed.2001.0905>